Generic Sign Rigidity of Real Projection Kernels from Two-Point Data

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Abstract

Let K be a real symmetric orthogonal projection. Its diagonal and the moduli of its off-diagonal entries are exactly the data contained in its principal minors of orders one and two, but they do not display the signs of K_{ij}. We prove that this apparent loss is generically repaired by idempotency. For every $N \geq 5$ and $2 \leq r \leq N-2$, almost every rank-r projection is determined, up to conjugation by a diagonal sign matrix, by $\{K_{ii}, |K_{ij}|\}$. The proof separates $K^2 = K$ into a diagonal modulus identity and a linear system for gauge-invariant triangle holonomies. A zero-subset-sum condition on the signed wedge weights is a finite certificate of uniqueness. We show that every forbidden subset sum is a nonzero real-analytic function on $\mathrm{Gr}(r, N)$; its zero locus has measure zero, and a finite union argument gives the theorem. We also give a certified reconstruction procedure: solve the finite holonomy system, test the certificate, fix a spanning-tree gauge, and recover all signs. The result is a theorem about finite real projection kernels. Its use for determinantal boundary data is only a dictionary: in a finite real pure determinantal completion, one- and two-boundary data generically fix the higher principal minors. No continuum, Hermitian, or quantum-gravity completeness claim is made.

1 Introduction

For a real determinantal point process, inclusion probabilities are principal minors of a symmetric contraction K [5, 6]. The order-one and order-two minors give

$$d_i = K_{ii}, \qquad \det K_{\{i,j\}} = d_i d_j - K_{ij}^2, \tag{1}$$

and hence determine $m_{ij} = |K_{ij}|$. They do not determine the signs. General principal-minor recovery therefore uses cycle data, normally supplied by minors of order at least three [1, 2, 8]. All principal minors are invariant under

$$K \longmapsto DKD, \qquad D = \mathrm{diag}(\epsilon_1, \ldots, \epsilon_N), \quad \epsilon_i \in \{\pm 1\}, \tag{2}$$

so recovery can never be sharper than this diagonal-sign gauge.

We add one hypothesis: K is an orthogonal projection. Projection kernels are basic extreme determinantal kernels [5, 7]; algebraically, the added input is simply $K^2 = K$. The question is whether this quadratic identity replaces missing cycle minors. Our answer is affirmative on a full-measure subset of every interior real Grassmannian covered below.

Theorem 1.1 (Generic projection sign rigidity)**.** *Fix $N \geq 5$ and $2 \leq r \leq N-2$. For almost every $K \in \mathrm{Gr}(r, N)$, if K' is a real symmetric projection satisfying*

$$K'_{ii} = K_{ii}, \qquad |K'_{ij}| = |K_{ij}| \quad (i \neq j),$$

then $K' = DKD$ for a diagonal sign matrix D.

Here "almost every" refers to the natural smooth measure class on $\mathrm{Gr}(r, N)$. The theorem includes generic full off-diagonal support; it is not a claim for every projection. The uniform statement is deliberately scoped to $N \geq 5$ and $2 \leq r \leq N-2$. Sparse supports, exceptional subset-sum loci, complex Hermitian projections, and continuum kernels require separate analyses.

The key observation is that the off-diagonal part of idempotency becomes *linear* after passing from edge signs to triangle holonomies. Sign rigidity is then a finite lattice intersection problem. This differs from the established principal-minor results: those recover cycle signs from higher minors, whereas here only orders one and two are given and idempotency supplies the extra equations.

2 Gauge-invariant form of idempotency

Let $K = K^\top = K^2 \in \mathbb{R}^{N\times N}$ have rank r. Put

$$d_i = K_{ii}, \qquad m_{ij} = |K_{ij}|, \qquad \sigma_{ij} = \operatorname{sgn} K_{ij} \quad (m_{ij} > 0). \tag{3}$$

For the main theorem the support is generically complete. On a complete support graph, define the triangle holonomy

$$h_{ikj} = \sigma_{ik}\sigma_{kj}\sigma_{ji} \in \{\pm 1\}. \tag{4}$$

It is invariant under (2). Conversely, a valid holonomy field determines the edge signs up to gauge: set the signs on a spanning tree, then use triangles to obtain every remaining edge. Valid fields obey the usual cocycle relations; not every arbitrary $\{\pm 1\}$ assignment to triangles is valid.

Proposition 2.1 (Idempotency split)**.** *For full off-diagonal support, $K^2 = K$ is equivalent to*

$$\sum_{k\neq i} m_{ik}^2 = d_i(1-d_i), \qquad i = 1, \dots, N, \tag{5}$$

$$\sum_{k\neq i,j} m_{ik}m_{kj}h_{ikj} = m_{ij}(1-d_i-d_j), \qquad 1 \leq i < j \leq N. \tag{6}$$

The first system contains only two-point data; the second is linear in h.

Proof. The diagonal equation $K_{ii} = \sum_k K_{ik}^2$ gives (5). For $i \neq j$, separating the $k = i, j$ terms in $K_{ij} = \sum_k K_{ik}K_{kj}$ gives

$$K_{ij}(1-d_i-d_j) = \sum_{k\neq i,j} K_{ik}K_{kj}.$$

Divide by σ_{ij} and use $\sigma_{ij}^{-1} = \sigma_{ji}$ to obtain (6). Reversing these steps recovers all entries of $K^2 = K$. □

Write (6) as $Ah = R$, where

$$A_{(ij),(ikj)} = m_{ik}m_{kj} \geq 0, \qquad R_{ij} = m_{ij}(1-d_i-d_j). \tag{7}$$

If two sign fields give projections with the same (d, m), their valid holonomies h, h' satisfy $A(h-h') = 0$. Thus the exact uniqueness condition is

$$\ker A \cap \{h - h' : h, h' \text{ valid holonomy fields}\} = \{0\}. \tag{8}$$

3 A finite uniqueness certificate

For a fixed projection, define the signed wedge weights in row (i, j) by

$$w_k^{(ij)} = m_{ik}m_{kj}h_{ikj} = K_{ik}K_{kj}\sigma_{ij}, \qquad k \neq i, j. \tag{9}$$

Lemma 3.1 (Zero-subset-sum certificate)**.** *Suppose K has full off-diagonal support and, for every edge (i, j), no nonempty subset of $\{w_k^{(ij)} : k \neq i, j\}$ has sum zero. Then every projection with the same diagonal and entrywise moduli is gauge-equivalent to K.*

Proof. Compare K with a rival sign field. Let S be the set of flipped edges. A triangle is flipped exactly when it contains an odd number of edges of S. Subtracting the two copies of (6) gives, in each row,

$$\sum_{k:\,(i,k,j) \text{ is flipped}} w_k^{(ij)} = 0.$$

The hypothesis makes every such subset empty, so no triangle is flipped. In the complete graph, triangles span the cycle space over $\mathbb{F}_2$. An edge set meeting every triangle evenly is therefore in the cut space. Hence S is a vertex cut, precisely the gauge transformation (2). □

This certificate is sufficient, not necessary. It can be checked exactly using meet-in-the-middle subset sums in $O(N^2 2^N)$ arithmetic operations. It is a *post-recovery* certificate: the weights use a candidate holonomy field. The data-only problem remains finite because one can solve (7) over valid cocycles (or enumerate a gauge-fixed edge-sign field), and then apply the certificate to each survivor.

4 Proof of generic rigidity

We now show that the certificate holds almost everywhere. For an edge (i, j) and a nonempty $T \subseteq [N] \setminus \{i, j\}$, define

$$F_{ij,T}(K) = \sum_{k \in T} K_{ik}K_{kj}. \tag{10}$$

Up to the common nonzero factor σ_{ij}, this is exactly the corresponding subset sum of (9).

Lemma 4.1 (Nonvanishing)**.** *For $N \geq 5$ and $2 \leq r \leq N - 2$, no function $F_{ij,T}$ is identically zero on* $\mathrm{Gr}(r, N)$.

Proof. Choose $k \in T$ and set $u = (e_i + e_j + e_k)/\sqrt{3}$. Since $r - 1 \leq N - 3$, choose distinct indices $a_1, \ldots, a_{r-1}$ outside $\{i, j, k\}$. Let K be the orthogonal projection onto

$$\mathrm{span}\{u, e_{a_1}, \ldots, e_{a_{r-1}}\}.$$

Then $K_{ik} = K_{kj} = 1/3$, while $K_{i\ell}K_{\ell j} = 0$ for every $\ell \neq k$. Consequently $F_{ij,T}(K) = 1/9$. The witness need not have full support; it only proves that the analytic function is nonzero. □

Proof of Theorem 1.1. The entries K_{ab} and the finitely many functions $F_{ij,T}$ are real analytic on the connected analytic manifold $\mathrm{Gr}(r, N)$. Each entry is not identically zero, so the union of all zero-entry loci has measure zero; outside it, K has full support. By Lemma 4.1, every $F_{ij,T}$ is nonzero as an analytic function. The zero set of a nonzero real-analytic function has measure zero. There are finitely many triples (i, j, T), hence almost every K is simultaneously full-support and free of zero subset sums. Lemma 3.1 then gives uniqueness up to gauge. □

The proof is qualitative but constructive enough to yield exact rational witnesses. If $V \in \mathbb{Q}^{r \times N}$ has full row rank, then $K = V^\top (VV^\top)^{-1} V$ is rational. Clearing denominators turns every certificate row into an integer subset-sum test. Such checks verify instances; the analytic argument above, not the finite sample, proves the almost-everywhere statement.

5 Certified reconstruction

Given exact (d, m) and the information that a real rank-r projection exists, a complete finite procedure is:

1. Check $\sum_i d_i = r$ and the modulus equations (5).
2. Fix a spanning-tree gauge (for complete support, take $\sigma_{1i} = +1$).
3. Solve (6) over valid $\mathbb{Z}_2$ cocycles. In the chosen gauge, $\sigma_{ij} = h_{1ij}$ for $i, j > 1$.
4. Reconstruct $K_{ii} = d_i$ and $K_{ij} = \sigma_{ij} m_{ij}$, then verify $K^2 = K$ exactly.
5. Apply Lemma 3.1. If it passes, the recovered gauge orbit is the unique one.

The method is exponential in its most direct form; Theorem 1.1 is an identifiability result, not a polynomial-time learning theorem. A useful local certificate can nevertheless decode one edge. If known legs give an anchored sum a and the sum of the absolute weights of all unknown legs is u, then $|a| > u$ fixes the sign of the full row. This dominance test certifies a warm-started solution but need not start from a tree gauge by itself.

6 Relation to principal-minor and boundary data

General symmetric-kernel recovery from principal minors is governed by signed cycles and principal-minor assignment [1–4, 8]. Those results explain why two-point data do not determine a generic symmetric kernel. Theorem 1.1 is compatible with them: it restricts the matrix to the projection variety, where idempotency supplies the missing cycle equations. Our proof does not establish stability under noisy data, sparse-support recovery, or the Hermitian phase analogue.

A subsequent related preprint by Villacis [9] uses corresponding principal minors of all orders for complex orthogonal projections to determine the coordinatewise absolute-value image of their ranges. The present theorem assumes only order-one and order-two data and restricts recovery to generic real projections up to diagonal-sign gauge. The different data requirements, domains, and recovery targets should be retained in any novelty comparison; this observation does not establish priority.

There is a limited gravitational application. In a finite real determinantal model, the n-point correlation function is the determinant of the $n \times n$ kernel restriction. Laplace-transformed one- and two-boundary data therefore determine d_i and m_{ij} in a finite discretisation. If the completion is also known to be a real projection, the theorem says that, generically, these data fix the kernel up to a gauge that leaves all principal minors unchanged. Higher correlators then follow. This is only a dictionary for a finite real pure determinantal completion. It does not prove that a gravitational path integral has such a completion, nor does it settle Airy-kernel continuum limits, complex phases, interactions, or ultraviolet completion.

7 Limitations and conclusion

The exceptional set is explicit but not classified: it is contained in the union of the entry-zero and subset-sum-zero analytic loci. Failure of the sufficient certificate does not imply a rival projection. Hermitian kernels replace $\mathbb{Z}_2$ holonomies by $U(1)$ phases, so the discrete subset-sum proof does not transfer. Continuum kernels require analytic control of nodal domains and integral, rather than finite, wedge sums.

Within its declared scope, the result is complete. The diagonal and entrywise moduli of a generic interior real projection determine its signs up to diagonal gauge. The proof reduces purity to a holonomy-linear system, supplies a finite certificate, and establishes that the certificate fails only on a measure-zero set.

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